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METHODS OF TEST FOR DAIRY INDUSTRY

PART V METHODS OF DAIRY PLANT CONTROL

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BUREAU OF INDIAN STANDARDS
MANAK BHAVAN, 9 BAHADUR SHAH ZAFAR MARG
NEW DELHI-110002

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Indian Standard

METHODS OF TEST FOR DAIRY INDUSTRY

PART V METHODS OF DAIRY PLANT CONTROL

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Indian Standard

METHODS OF TEST FOR DAIRY INDUSTRY

PART V METHODS OF DAIRY PLANT CONTROL

0. FOREWORD

0.1 This Indian Standard (Part V) was adopted by the Indian Standards Institution on 21 December 1962, after the draft finalized by the Dairy Industry Sectional Committee had been approved by the Agricultural and Food Products Division Council.

0.2 The Part V of the standard covers the methods commonly used for control work in dairy plants. The other parts of this standard cover the following:

Part I	Rapid Examination of Milk
Part II	Chemical Analysis of Milk
Part III	Bacteriological Analysis of Milk
Part IV	Determination of Freezing Point Depression of Milk by Hortvet Method

0.3 The adoption of this part will help to achieve uniformity in the methods of dairy plant control, thereby facilitating the interpretation and comparison of results.

0.4 Wherever a reference to any Indian Standard appears in this standard, it shall be taken as a reference to the latest version of the standard, with amendments, if any.

0.5 In the formulation of the Part V of the standard, considerable assistance has been derived from the following publications:

DAVIS, J. G. Laboratory Control of Dairy Plant. London. Dairy Industries Ltd, 1956.

STANDARD METHODS FOR EXAMINATION OF DAIRY PRODUCTS, 11th ed. American Public Health Association, 1960.

LABORATORY MANUAL. Washington. Milk Industry Foundation, 1959.

Full use has been made of the valuable information received from the National Dairy Research Institute, Karnal.

0.6 Metric system has been adopted in India and all quantities and dimensions in this standard have been given in this system.

0.7 In reporting the result of a test or analysis made in accordance with this standard, if the final value, observed or calculated, is to be rounded off, it shall be done in accordance with IS : 2-1960 Rules for Rounding Off Numerical Values (*Revised*).

1. SCOPE

1.1 This standard specifies methods generally used for ancillary tests commonly employed in dairy plants. The specific methods to be employed would depend upon the object of analysis.

2. QUALITY OF REAGENTS

2.1 Unless specified otherwise, pure chemicals and distilled water [*see* IS : 1070-1960 Specification for Water, Distilled Quality (*Revised*)] shall be employed in tests.

NOTE — 'Pure chemicals' shall mean chemicals that do not contain impurities which affect the experimental results.

3. CHECKING HOLDING TIME OF HTST PASTEURIZERS

3.1 The holding time of a high-temperature short-time (HTST) pasteurizer is the flow time of the fastest particle of milk at the prescribed temperature through the holder section, which may either be a pipe (single or nested) or a series of plates. The checking of holding time of the pasteurizing equipment is particularly important with the HTST plant where, at certain temperatures, a reduction of a few seconds in that time may have a serious effect on the quality of the finished product. Any method which is serviceable for this purpose ought to give a reasonably accurate measurement of the time in which any one particle of milk can pass through the holder section or plates. This section should have necessary fittings at the beginning as well as at the end for purposes of checking the accuracy of holding time, which can be carried out by chemical methods.

3.2 Chemical Methods

3.2.1 *The Dye-Injection Method* — Inject methylene blue, into the milk inlet to the holding plate, or pipe while the plant is being run on water. Start simultaneously a stop-watch. After the lapse of, say, 10 seconds, collect samples numbered 1, 2, 3, etc, in ten test-tubes at intervals of one second from a special cock at the outlet from the holding section, and place the tubes in their numerical order in a rack. If the sampling was commenced at 10 seconds after the injection of the dye, and if the colour first appears in, for example, No. 6 tube, No. 5 tube being quite clear, then the holding time shall be taken as $5 + 10 = 15$ seconds.

3.2.2 *Dummett and Mongar Method*

3.2.2.1 *Equipment*

a) *Spring loaded syringe or gun* — with lock and release device at top and with injection spray nozzle having 6 ports to screw into the holder.

b) *Fitting on inlet end of holding tube* — to receive the spray nozzle.

c) *Pipettes* — for filling the gun.

d) *Sampling valve fitting* — at discharge end of holding section.

e) *Test-tubes* — of 100-ml capacity marked at the level of 76 ml.

f) *Rack for test-tubes*

g) *Glass-stoppered bottles* — of amber colour.

NOTE — The spring loaded syringe or gun used in this test is adjusted to provide a pressure of 7·03 kg/cm² which is greater than the pressures likely to be encountered in holding section. This is essential for accuracy. The spray nozzle with its 6 ports is mounted in the fitting so as to direct the injection at a right angle to the line of flow in the holder, thus affording a sharp front of the nickel chloride solution when it enters the holding section. This too is essential for accuracy.

3.2.2.2 *Reagents*

a) *Nickel chloride solution* — saturated solution prepared by dissolving 40 g nickel chloride in 60 ml of water. Store the solution in an amber coloured glass-stoppered bottle.

b) *Dimethylglyoxime solution* — Dissolve one gram of dimethylglyoxime in 100 ml of ethyl alcohol, 95 percent by volume, and add to this solution one millilitre of ammonium hydroxide (sp gr 0·90). Store in an amber coloured glass-stoppered bottle.

3.2.2.3 *Procedure* — Screw the injection nozzle and sampling valve into suitable fittings. Run the pasteurizer on water under normal conditions at the rated flow, which should be checked. Open the sampling cock at the outlet end of the holding section and adjust it so that it delivers 75 ml per second. Add about 1 ml of the dimethylglyoxime solution to each of the 10 test-tubes (*see* **3.2.1**) held in a rack. Load the gun by pulling the plunger back against the spring, until the stop lugs are above the guide slots, then twist the handle slightly to lock the plunger in position against the compression of the spring. Fill the gun with the saturated nickel chloride solution by means of a pipette, taking care that no air is entrapped. Inject the nickel chloride solution by releasing the lock on the handle and at the same time start the stop-watch. As the stop-watch reads 10 seconds, pass the rack of tubes under the stream, filling each tube to the mark in succession. Count the number of colourless tubes. This number plus 10 gives the holding time of the unit. The positive reaction gives a yellow colour or pink precipitate depending upon the concentration of the reactants.

Repeat the test twice again and determine the holding time by averaging the results of the three tests. If none of the tubes shows colour, repeat the test by starting the testing 15 seconds after injection of the nickel chloride solution.

NOTE 1 — Wash the tubes carefully after each test with suitable brush and clean water. Do not allow any of the reagents to stain the rack or the hands as this may lead to false positive tests. Stains on the rack may be removed with dilute hydrochloric acid.

NOTE 2 — The pasteurizer should be operated for 5 minutes at full capacity between tests to ensure that it is cleared of the injected solution.

3.2.3 *Sodium Nitrite Method*

3.2.3.1 *Equipment* — The equipment required is the same as described under **3.2.2.1**.

3.2.3.2 *Reagents*

a) *Sodium nitrite solution* — Dissolve 10 g of sodium nitrite in 100 ml of water. Store in amber coloured glass-stoppered bottle.

b) *Griess-Ilosvay reagent*

Dissolve 0·5 g of sulfanilic acid in 30 ml of glacial acetic acid and 120 ml of water.

Dissolve 0·1 g of α-napthylamine in 120 ml of boiling distilled water. Cool the solution and add 30 ml of glacial acetic acid and filter.

Combine the two solutions and store in an amber coloured glass-stoppered bottle.

3.2.3.3 *Procedure* — Proceed as in **3.2.2.3**, except that the sodium nitrite solution is injected into the holding section, and the Griess-Ilosvay reagent is placed in the test-tubes. A positive test shows a pink colour.

3.3 Indication

3.3.1 The importance of sensitivity in checking the holding time lies in the fact that not all the particles of milk in the holding section pass through it at the same rate. Those in the centre of the stream will travel more quickly than those away from the centre. Ideally it is the speed of the fastest particle which is necessary to be measured and the sensitivity of the test depends on how nearly this object is achieved.

3.3.2 The holding time in the HTST pasteurizer is a function of the rate of milk flowing through it and varies according to the output of the timing pump. Therefore, once the holding time has been established for the particular flow-rate in a given plant by any one of the above described methods, thereafter, throughout the life of that plant (provided no alterations have been made to its holding section), a holding time test need be no more than a simple flow-rate test. If the flow-rate remains constant, then the holding time will also be constant. If in the test, the flow-rate is found to have altered, then the holding time test shall be re-applied. In a properly managed plant, it is customary that once the speed of the timing pump has been adjusted and the holding time test shows the stipulated holding time, a seal is attached in such a way that the timing pump cannot be operated at a higher speed without first breaking the seal.

4. ASSESSING THE CLEANLINESS AND STERILITY OF PLANT

4.1 Methods of measuring the extent of residual bacterial contamination on an equipment in a dairy plant fall into the following main groups:

a) those in which a given area or a given item is either swabbed, or rinsed with a known volume of Ringer's solution or phosphate buffer;

b) those in which a contact agar colony count is obtained; and

c) those in which the product which passes through or over the equipment is measured for its bacteriological quality and especially for changes in quality.

4.1.1 Swab methods may be applied to large or small areas on flat or curved surfaces while rinsing methods are particularly applicable to bottles, cans and other utensils which could be closed for rinsing. The methods in the first two groups are more direct, in the sense that an attempt is made to measure the actual number of micro-organisms which remain on the plant and to demonstrate their growth by allowing the bacterial cells to form colonies. The methods are, however, rather expensive, as all methods involving colony counts are bound to be. The methods in the third group are much simpler and cheaper as all they require is some assessment of bacteriological quality, which can very easily be made by the resazurin or methylene blue test. Further, such methods have the added advantage that the effect of contamination on the keeping quality of the product can be directly demonstrated.

4.2 Swab Method

4.2.1 The swab consists of cotton wool or other similar material which is bound together at the end of a metal wire, preferably of stainless steel. The other end of the wire is commonly looped and the swab is immersed in a quarter strength Ringer's, solution or phosphate buffer [*see* **5.2.6** of IS : 1479 (Part III)-1962]. The tube is plugged with cotton and then boiled. The swab and the solution are then sterilized by autoclaving or intermittent steaming in the usual way. A suitable allowance for evaporation of the solution can be made by finding out in practice what loss in volume occurs as a result of the sterilization process and adding that amount initially. As there is a possibility of a large error in the method, there is no point in striving for great accuracy in measuring the volume of the solution, and it can be assumed that changes in strength are of no importance.

4.2.1.1 Probably the biggest disadvantage of the swab method is that the proportion of micro-organisms removed from the surface will vary according to the method of swabbing and, further, an unknown proportion of those organisms removed will remain entangled in the swab and thus not be released into the solution when a portion of this is plated. Contact agar methods, nearly always, give an appreciably higher count. The outstanding advantage of the swab method is that it can be used for any type of plant, including the most inaccessible corners and crevices. In using the swab method, it is always best to swab that part of the equipment which is known by experience to be the most difficult to clean and sterilize. Thus, there is no point in swabbing portions of a smooth surface for example, the side of a tank, and ignoring such items as cocks, valves, bends in pipelines, etc.

4.2.1.2 In order to obtain a quantitative assessment, some laboratories swab a given area, but for the reasons given above, this is not always possible. It is the amount of overall contamination which counts rather than the extent of contamination in selected places. As it is impossible to swab a given area for such items as cocks, filler valves, etc, there is much to be said for swabbing one entire item of equipment, for example the interior of a cock, pump, a filler valve, etc, and there is no great advantage to be gained by trying to express results as the number of bacteria for a given area, rather than for an individual item.

4.2.1.3 The swab method given in **4.2.2** is intended for assessing the bacteriological condition of equipment used in handling milk where the rinsing method is not applicable. Such an equipment will include tanks of various sizes, all types of coolers, pipe-lines as well as small incidental fittings, such as cocks, agitators, air vents, etc. In testing any pieces of equipment, more than one area should be swabbed, particularly where there are points difficult of access for cleaning.

4.2.2 *Apparatus*

a) 25 × 2·5 cm test-tubes rimless, of heat-resistant glass.

b) 35 cm of 2·6-mm stainless steel wire — formed into a loop at one end, leaving a straight length 30 cm long, and notched at the other end to hold the ribbon gauze.

c) 50-mm unmedicated ribbon gauze.

4.2.3 *Preparation of Swab* — The swab shall be 50 mm in length and shall consist of 150 mm of the gauze wound round the notched end of the stainless steel wire and secured with thread.

4.2.4 *Sterilization of Swab* — Place the swab in 25 ml of the Ringer's or phosphate buffer solution in a test-tube, plug with cotton wool, cover the plug with grease-proof or similar paper and sterilize by autoclaving at 120°C for 15 minutes.

4.2.5 *Swabbing* — Where possible an area of 900 cm² shall be examined. Press the swab with a rolling motion against the side of the glass tube to remove the excess liquid. Remove the swab and with heavy pressure rub back and forth over the area to be examined so that all parts of the surface are treated twice. The swab shall be rotated so that all parts of it make contact with the surface. Return the swab to the tube.

4.2.6 *Treatment of Samples* — If swab samples are not tested immediately, they shall be placed at once in a refrigerator at 2° to 3°C and examined not later than 24 hours.

4.2.7 *Testing Swab* — After not less than 5 minutes contact of the swab and the liquid, mix by swirling the swab vigorously in the solution 6 times. Remove the swab, taking care to express the excess liquid by pressing against the side of the tube. After thorough mixing by rotation between the palms of the hands, prepare 1 ml and 0·1 ml plates, using yeastrel milk agar or tryptone glucose agar and incubate at 37·0° ± 0·5°C for 48 hours before counting [*see* **5** of IS : 1479 (Part III)-1962].

4.2.8 *Recording Results* — The results shall be calculated and recorded as the colony count per 900 cm² (colony count per ml × 25 × area factor, if not 900 cm²).

4.2.9 *Interpretation of Results* — The following standards are suggested as a guide for grading:

Colony Count per 900 cm²	*Classification*
Not more than 5 000	Satisfactory (*S*)
Over 5 000 to 25 000	Fairly satisfactory (*FS*)
Over 25 000	Unsatisfactory (*US*)

NOTE 1 — When hypochlorite solution has been used for the sterilization of any surface, crystalline sodium thiosulphate shall be added to the Ringer's solution or phosphate buffer before autoclaving so as to give a concentration of approximately 0·05 percent. Where quaternary ammonium compounds have been used, add 0·4 percent lecithin and 1·0 percent Tween 80 (polyoxyethylene sorbitan monooleate) or Tween 20 (polyoxyethylene sorbitan mono-laurate) to the Ringer's solution or phosphate buffer.

NOTE 2 — The area of swabbing is to be calculated approximately for pipes and cocks, and if, as is possible, the swab is not long enough to traverse the required area, a known fraction of it shall be treated and the appropriate factor used in the final calculation. A mask 12·5 × 7·5 cm made from a thin sheet of stainless steel, aluminium, etc, with a cut out area of 10 × 5 cm in the centre may be used to facilitate swabbing any required area. The masks can be sterilized either by autoclaving at 120°C for 15 minutes or by heating at 160°C for one hour.

NOTE 3 — The stainless steel wire has a tendency to rust when kept in Ringer's solution or phosphate buffer. To overcome this, it should be thoroughly cleaned with abrasive powder, allowed to stand in concentrated hydrochloric acid for about 15 minutes and then washed with distilled water.

4.3 Rinse Method

4.3.1 The rinse method consists simply in rinsing under standard conditions any one container. It is thus simpler than the swab method because no swabs are required. It has the disadvantages that a larger volume of Ringer's solution or phosphate buffer is usually required and the method can, of course, only be used where rinsing is feasible for containers, such as bottles and cans, and possibly for tipping tanks, tankers, etc. The rinse method is almost universally used for milk bottles and milk cans.

4.3.1.1 *Milk rinse method* — Determine sterility of pipe lines, coolers and bottle fillers by removing sample of milk (*A*) with sterile pipette or dipper directly from pasteurizing vat just before starting the milk pump, and withdraw another milk sample (*B*) from one of the bottle-filler valves before bottling into sterile containers. Test both the samples for standard plate count and presence of coliform organisms according to procedures described in IS : 1479 (Part III)-1962.

4.3.1.2 *Interpretation* — The following interpretation is suggested as a guide:

a) When standard plate counts of sample (*B*) exceed those of sample (*A*) by 100 percent plus 2 000, sanitization of equipment is considered unsatisfactory.

b) Since samples from pasteurizers should normally be free from coliform bacteria, positive coliform tests on sample (*B*) indicate unsatisfactory sanitation or recontamination of some portion of equipment through which milk has passed.

5. ASSESSING STERILITY OF MILK BOTTLES

5.1 Rinse Method — The method outlined is designed to give information on the bacteriological condition of bottles immediately after washing. The minimum number of bottles to be examined at each test shall be four selected at random. Where a machine is giving unsatisfactory results, it may be necessary to test a complete row of bottles.

Select bottles for examination immediately after washing, close with a sterile rubber bung (*see* Note) and keep at once embedded in ice or in a refrigerator at 2° to 4°C. Test within 24 hours. Bottles shall be examined as soon after sampling as possible. This is particularly important where hydrochlorites are used in the final rinse.

NOTE — The rubber bungs shall be wrapped in grease-proof paper and sterilized in the autoclave as 120°C for 15 minutes. Before closing the bottle with the bung, remove the grease-proof paper and use it to cover the bung and neck of the bottle, being held in position by a rubber band.

5.1.1 *Rinsing* — Add 20 ml of the sterile solution (*see* **4.2.1**) to the bottle and replace the bung. Hold the bottle horizontally in the hands and rotate gently 12 times in one direction so that the whole of the internal surface is thoroughly wetted. Allow the bottle to stand for not less than 15 and not more than 30 minutes and again gently rotate 12 times so that the whole of the internal surface is thoroughly wetted.

NOTE 1 — Twenty millimetres of the solution shall be used irrespective of the size of the bottle. Where bottles are taken from a hot section of a machine for special purposes, they shall be fitted with a bung and allowed to cool before rinsing. Where the final rinse water is known to be chlorinated, crystalline sodium thiosulphate shall be added to the Ringer's solution before autoclaving to give a concentration of 0·05 percent.

NOTE 2 — No mention has been made of the question of wetness of bottles but if they are washed properly and sufficient time allowed for drainage, little moisture should be left in them. Obviously the presence of excessive moisture is undesirable and calls for attention on the part of the management.

5.1.2 *Testing* — Immediately prepare 5-ml plates in duplicate using yeastrel milk agar or tryptone glucose agar and incubate at 37·0° ± 0·5°C for 48 hours before counting.

NOTE — Additional information may be obtained by incubating one set of plates at 22°C and by testing for the presence of coliform organisms and for milk souring organisms.

5.1.3 *Recording Results* — The results shall be recorded as the colony count per bottle. That is the sum of the counts on the two plates multiplied by two.

5.1.4 *Interpretation of Results* — One colony per millilitre of the capacity of the bottle is satisfactory and over one colony per millilitre unsatisfactory.

5.2 Agar Roll Method — This method is very simple in operation and avoids requirements for dilution water, pipettes, petri-dishes, etc, and hence assists the worker to carry out a larger number of tests in a short time. The bottles for examination shall be collected in the same manner as described under **5.1**. Keep the bottles for half an hour in an airconditioned room or for 10 to 15 minutes in a refrigerator. This will cool the glass and help the formation of an even film of agar when the bottle is spun as stated in **5.2.1**.

5.2.1 *Testing* — Pour 10 to 15 ml (depending upon the size of the bottle) of melted and cooled yeastrel milk agar or tryptone glucose extract agar into the bottle, replace the rubber bung and then spin it rapidly either mechanically or manually so that an even film of agar is formed on the inner surface of the bottle. Incubate the bottle at 37·0° ± 0·5°C with the stopper downwards so that any condensate drains towards the stopper. After incubation, count in a bright light the number of colonies developed in the bottle.

5.2.2 *Recording of Results* — The results shall be recorded as the colony count per bottle.

5.2.3 *Interpretation of Results* — The results are interpreted in the same manner as in **5.1.4**.

NOTE — This method for assessing the sterility of milk bottles shall be used with caution when the final rinse water in the bottle washer is chlorinated beyond 7 parts per million.

6. ASSESSING STERILITY OF MILK CANS

6.1 The method is designed to give information on the sterility condition of milk cans. The minimum number of cans to be examined at each test shall be four, selected at random. These cans shall be examined by the rinsing method. But a larger number shall be subjected to visual examination. Cans containing milky water or easily removable milk solids as distinct from film or scale shall be reported as unsatisfactory without testing.

6.2 Selection of Cans — Cans for examination shall be selected immediately after washing, and examined within an interval of not less than â half and not more than one hour.

6.3 Visual Examination — The general conditions of the cans shall be noted, paying particular attention to the following points:

a) bad dents, rusting of inner surfaces, open seams, poor lids;

b) presence or absence of film, scale or milk solids;

c) wetness of cans. The degree of wetness shall be recorded as:

Dry — if there is no visible moisture in the can;

Moist — if beads of moisture, but no visible pool, is present at the bottom of the can; and

Wet — if there is obvious pool of water at the bottom of the can.

6.4 Rinsing — Pour 500 ml of the sterile solution [*see* **5.2.6** of **IS** : 1479 (Part III)-1962] over the inside of the lid into the can. Replace the lid, lay the can on its side and roll to and fro so that it makes 12 complete revolutions. Sample the rinse by pouring direct from the can into a sterile container.

6.4.1 If desired, the can and lid may be examined separately.

6.5 Treatment of Rinse Samples — If the rinse samples are not tested immediately, they shall be placed at once in a refrigerator at 2° to 3°C and examined not later than 24 hours.

6.6 Testing Rinse Samples — Mix the sample by inverting the container slowly three times. Pour 1- and 0·1-ml plates, using milk yeastrel agar, and incubate at 37·0° ± 0·5°C for 48 hours before counting. The results shall be recorded as the colony count per can (colonies per millilitre of rinse × 500). Calculate the colony count per litre capacity of the can.

6.7 Interpretation of Results — The following interpretation is suggested:

Grading *Colony Count per Litre Capacity*	Classification (or Dry or Moist Cans)
Not more than 1 000	Satisfactory
More than 1 000 but less than 5 000	Fairly satisfactory
Above 5 000	Unsatisfactory

7. EXAMINATION OF DETERGENTS

7.1 Types — The term 'detergent' is now used for a wide variety of chemical substances. These fall into two clearly defined groups — the inorganic and the organic. The constituents of proprietary detergents may be any of the following:

a) *Alkalis*

1) Caustic soda or sodium hydroxide, NaOH
2) Soda ash or sodium carbonate, Na_2CO_3
3) Sodium metasilicate, Na_2SiO_3
4) Trisodium phosphate, Na_3PO_4

b) *Polyphosphates (for Water Softening)*

1) Sodium hexametaphosphate, formula usually given as $Na_6(PO_3)_6$.
2) Tetra-sodium pyrophosphate, $Na_4P_2O_7$.

c) *Synthetics*

1) *Anionic*

i) Sulphated primary alcohols (for example, Lissapol C)
ii) Sulphated secondary alcohols (for example, Teepol)
iii) Alkyl aryl sulphonates (for example, Acinol)
iv) Sulphosuccinic esters (for example, Alcopol)

2) *Cationic* — Quaternary ammonium compounds:
   i) Single long chain (for example, Cetavlon)
   ii) Benzyl (for example, Roccal)
   iii) Twin short chain (for example, Deciquam)
   iv) Pyridinium (for example, C.P.C. Liquid)

3) *Non-ionics*

   Various ethers and esters of the type RO $(CH_2CH_2O)_n$ H, where R may be aliphatic and/or hydrocarbon residue (for example, Lissapol N).

7.1.1 In addition to these, some proprietary mixtures may contain colloidal clays (for emulsification), sulphite (to protect tin), sodium metasilicate (to protect aluminium), soaps, and other substances. Anionic and cationic compounds are not usually found together as they often form an insoluble compound when mixed. Some proprietary detergents contain hypochlorites.

7.1.2 Usually detergent preparations consist of the following:

Purpose Used	*Type of Detergent*
Machine bottle washing (jet) (*J*)	Alkalis only
Machine bottle washing (soaker) (*S*)	Alkalis with synthetics
Hand bottle washing (*H*)	Sodium carbonate with synthetics
Can washing (jet) (*C*)	Alkalis only

7.1.2.1 There are exceptions to the above generalization. *J* and *C* types sometimes contain synthetics but extreme care has to be taken to avoid foaming. *S* types are sometimes nearly all or entirely caustic soda.

7.2 Examination — For the examination of an unknown mixture of detergents the following simple scheme is useful for obtaining an approximate idea of the detergent.

7.2.1 Prepare a one percent solution (*w/v*) in water and examine by the following tests:

a) *p*H — *J* and *S* types (*see* **7.1.2**) are usually strongly alkaline showing a *p*H of 12 to 13. *H* type is less alkaline, having *p*H 10 to 12. Pure one percent solutions of the chief detergent constituents have *p*H values approximately as follows:

Sodium hydroxide, NaOH	13·1
Sodium metasilicate, Na_2SiO_3	12·5
Sodium carbonate, Na_2CO_3	11·4
Trisodium phosphate, Na_3PO_4	12·0

Sodium sesquicarbonate, $Na_2CO_3.NaHCO_3$	9·9
Sodium bicarbonate, $NaHCO_3$	8·4
Sodium hexametaphosphate, $Na_6(PO_3)_6$	7·0

NOTE — For the determination of *p*H the 'alkali' glass electrode with any standard *p*H meter can be used. For determining *p*H colorimetrically use a suitable comparator disc and indicator.

b) *Carbonates* — Add dilute hydrochloric acid (1 : 3). Immediate effervescence indicates carbonates.

c) *Phosphates* — Add dilute nitric acid (1 : 3), boil for 3 minutes and cool. Add ammonium molybdate solution 5 percent (*w/v*) and warm to 60°C. Yellow precipitate indicates phosphates.

d) *Silicates* — Filter the solution from test (c) above. Add a pinch of oxalic acid. Add a drop of 0·05 percent benzidine (*w/v*) and 5 ml sodium acetate solution 5 percent (*w/v*). A blue colour indicates silicates.

e) *Soaps* — Acidify to make red to methyl red and boil. Separation of fatty acids (layer on top) indicates soaps. (Silica may also separate in this test.)

f) *Synthetics* — Remove fatty acids from test (e) above by filtration. Shake solution violently. A froth indicates synthetics.

g) *Cellulose ethers* — Boil some of the solution from test (f) above. A gel may indicate cellulose ethers; separation of an oil indicates polyethylene glycols.

h) *Cationic or non-ionic compounds* — If test (f) above is positive, add 3 ml of chloroform and a few drops of one percent methylene blue solution to the solution (e) above and shake vigorously. A blue chloroform layer indicates a sulphated or sulphonated product. A colourless chloroform layer indicates a cationic or non-ionic compound.

j) *Quaternary base* — Add about 3 ml of chloroform, 0·5 ml of sodium chloride and a few drops of bromophenol blue (0·1 g in 100 ml rectified spirit to the test solution (f). A blue chloroform layer indicates a quaternary base.

k) *Polyethylene glycol* — To 5 ml of reagent (174 g ammonium thiocyanate and 2·8 g cobalt nitrate in one litre) add 5 ml of test solution (f) above and shake vigorously. Stand for 2 hours. If the colour remains blue-violet, a polyethylene glycol is indicated.

7.3 Test for Efficiency of Dairy Detergents — A satisfactory detergent, when used under the correct conditions of temperature and concentration, should (a) remove milk fats and other deposits, (b) be easily removed by rinsing water, (c) not unduly corrode the metal surface, (d) not cause an undue deposit of scale when used in hard water, and (e) leave the surface in a reasonably good bacteriological condition so that subsequent sterilization should render it almost sterile. In order to judge the efficiency of dairy detergents from the above points of view, they are usually tested for their solubility, *p*H, wetting ability, emulsifying power, cleaning efficiency, rinsibility, general bactericidal property and corrosive action. For carrying out the different tests the solutions of the detergent materials shall be prepared in distilled water in the concentrations generally recommended for actual application (0·5 to 2 percent). For finding the solubility, however, the material should be added to water in a concentration of 10 percent.

7.3.1 *Solubility* — Add 10 g of the detergent to 100 ml of water at 30°C and filter the solution. Weigh the residue after drying an aliquot of the filtrate. On the basis of the amount of material dissolved in water, the detergent may be graded for solubility as follows:

Amount of Material Dissolved in 100 ml Water	*Grade*
More than 5 g	Good
2 to 5 g	Fair
Less than 2 g	Poor

7.3.2 *pH of the Material* — Take 10 ml of the detergent solution in a test-tube and add suitable indicator strip [*see* **7.0** of IS : 1479 (Part I)-1960]. From the change in colour of the strip, the approximate *p*H can be found. Alternatively, the *p*H can be determined electrometrically.

7.3.3 *Wetting Property* — Measure the time required for a skein of unboiled grey 2-ply cotton yarn weighing 5 g to sink when placed in 500 ml of the detergent solution as used in a 500-ml measuring cylinder, the skein carrying a 1·5-g sinker. Hold just above the surface of the solution by attaching to it a heavier weight called the anchor, which weighs at least 20 g. Before starting the experiment, release the anchor gently and allow the skein to be wetted and to sink down. The experiment is carried out at room temperature. Note the time taken for the skein to sink to the bottom. The detergent may be graded for wetting property as follows:

Time Taken to Sink to the Bottom	*Grade*
Less than 1 minute	Good
1 to 5 minutes	Fair
More than 5 minutes	Poor

7.3.4 *Emulsifying Power* — To a 100-ml portion of the detergent solution contained in a tall cylinder with a glass stopper, add 1·0 ml of melted butter-fat. Shake the solution and observe the length of time for which the emulsion persists. The following criteria are used for grading the materials for their emulsifying power:

Period of Persistence of Emulsion	*Grade*
More than 60 minutes	Good
15 to 60 minutes	Fair
Less than 15 minutes	Poor

7.3.5 *Cleaning Efficiency* — The cleaning efficiency of the detergents is determined on artificially soiled bottles.

7.3.5.1 Pour 5 ml of milk into each bottle and allow the bottles to dry. While drying, tilt the bottles occasionally so as to soil the bottles uniformly. Pour 100 ml of the detergent solution into the bottle and rotate the bottle in a horizontal position in order to make the solution come in contact with all portions of the bottle. Shake the bottle vigorously for about 25 to 30 minutes by up-and-down movements over a distance of 30 cm. Pour out the detergent solution along with the milk residue. Allow the bottle to drain for 30 minutes in a slanting position. In the case of detergent materials, which cause plenty of foaming, it is necessary to rinse the bottles with a small quantity of water to remove the residual foam.

7.3.5.2 Assess the cleaning efficiency on the basis of visual observation for the complete removal of milk particles and microdroplets of water from the surface as well as the absence of any foggy appearance. The presence of milk particles adhering to the surface is confirmed by shaking with a solution of dilute gentian violet and noting the violet stained spots.

7.3.6 *Rinisibility* — Moisten the fingers with the solution and then rub the fingers gently twice per second under a tap which is regulated to give a stream of 1·25 cm in diameter. Count the number of seconds required for the elimination of soapy feel on the fingers. Take the average of three experiments in each case to indicate the rinsibility.

The following standards are recommended for classifying the materials for rinsing efficiency:

Time Required for the Removal of Soapy Feel	*Grade*
Less than 3 seconds	Good
3 to 5 seconds	Fair
More than 5 seconds	Poor

7.3.7 *General Bactericidal Property* — Prepare the detergent solution in the concentration recommended by the manufacturer. Also prepare a 0·5 percent solution of sodium hydroxide in distilled water. Take 10-ml quantities of the two solutions in sterile test-tubes (15 cm × 1·6 cm) and sterilize at 108°C for 5 minutes. Cool the tubes to 50°C and place in a water-bath at this temperature. One millilitre of a 24-hour old milk culture of a strain of *Escherichia coli* is added to each tube by means of a sterile pipette. Mix the contents of the tubes by rotating with the palm and allow the tubes to stand in the water-bath. Keep the contents of the tube mixed by shaking at intervals of one minute. After 5 minutes draw one millilitre of the solution containing the culture from each tube and transfer to 9 ml of saline solution and then plate out using tryptone dextrose agar [*see* **5.3.1** of IS : 1479 (Part III)-1962]. Incubate the plate at 37·0°C ± 0·5°C for 48 hours and count the number of colonies of surviving organisms. The bactericidal power of the detergent is assessed on the basis of the percentage destruction of organisms compared to that observed in the case of cells exposed to the action of sodium hydroxide solution.

7.3.8 *Corrosive Action on Metals* — Take rectangular test pieces of stainless steel (Grade 07Cr19Ni9 of IS : 1570-1960 Schedules for Wrought Steels for General Engineering Purposes), mild steel heavily tinned by hot-dip process and aluminium alloy NS3 conforming to IS : 737-1955 Specification for Wrought Aluminium and Aluminium Alloys, Sheet and Strip (for General Engineering Purposes) and degrease with benzene. Suspend the test pieces in boiling 2N sulphuric acid for 2 minutes. Wash the strips thoroughly with distilled water, dry and store in a desiccator. The initial weights of the strips are taken accurately in a chemical balance. Take six lipless beakers of 250-ml capacity. Into three beakers pour 200 ml of 0·5 percent solution of caustic soda. Into the other three beakers the detergent solution (prepared in boiled and cooled glass-distilled water) under test is introduced. Hold the test pieces in a slanting position with at least 5 cm of each strip immersed in the solutions. Cover the beakers with watch-glasses and keep at a temperature of 60°C in an electrically heated oven for 2 hours. Remove the test pieces, rinse with distilled water, dry and re-weigh. If the mean of the losses in weight of the three test pieces in the detergent solution exceeds that of the test pieces suspended in sodium hydroxide solution, the corrosive action of the detergent solution is considered to be strong. If the mean of the losses in the former case is less than that in the latter, the corrosion is taken to be weak. Examine the strips, also for any discolouration, dulling, etching or pitting.

8. DETERMINATION OF STRENGTH OF WASHING SOLUTIONS

8.1 The strength of the detergent solution used depends on the use for which it is required. The strength of the washing solution is expressed in terms of sodium hydroxide (NaOH) but because of its caustic nature, sodium hydroxide is not often used direct especially at higher concentrations. An alkalinity equivalent to 0·5 percent caustic soda (*w/v*) is generally used for washing bottles and milk cans. For dairy pipelines, valves and fittings, 1·5 percent alkali (*w/v*) solution is used, while for cleaning heat exchangers when there is a likelihood of dried milk residues being left, 2·5 percent solution (*w/v*) is used. The efficiency of detergent solutions is increased when used hot (50° to 70°C).

8.1.1 The strength of the alkali in bottle washers has to be checked daily. One test shall be made at the beginning of the run after the alkali charge has been added and properly mixed. A second sample should be tested at the completion of the bottle washing operation. The alkali solution in can-washer tank should be checked at the beginning and the end of the can-washing operation and as often in between as is necessary, to assure proper strength of the solution during the operation.

8.1.2 For determining the strength of detergent solutions either titration methods or measurement of *p*H changes may be employed. For measuring *p*H the indicator methods are not so suitable due to the instability of these indicators at high *p*H (10 to 12). On the other hand, potentiometric method is quite suitable provided glass electrode is used. The titration method in various modifications is commonly used on account of its simplicity and low cost.

8.2 Rapid Test (Qualitative)

8.2.1 *Reagents*

8.2.1.1 *Standard hydrochloric acid* — 0·1 N or 0·3 N or 0·5 N, depending upon the strength of detergent to be tested, accurately prepared.

8.2.1.2 *Phenolphthalein indicator solution* — Dissolve one gram of phenolphthalein in 100 ml of 95 percent ethyl alcohol by volume. Add 0·1 N sodium hydroxide solution until one drop gives a faint pink colouration. Dilute with distilled water to 200 ml.

8.2.2 *Procedure* — Take 10 ml of the detergent in an Erlenmeyer flask and add 12·5 ml of 0·1 N standard hydrochloric acid, if the detergent to be tested is equivalent to 0·5 percent caustic soda. Add a few drops of the phenolphthalein indicator solution. If the mixture turns red, it indicates that the detergent solution has an alkalinity greater than 0·5 percent caustic soda (*w/v*), and is of sufficient strength. If, on the other hand, the mixture remains colourless, it is an indication that the solution is too weak and that more detergent needs to be added. For examining 1·5 and 2·5 percent detergent solutions, 0·3 N and 0·5 N hydrochloric acid shall be used and the above procedure followed.

8.3 Rapid Test (Quantitative) — A convenient method for determining the strength of alkali solutions is to prepare a standard acid solution of such strength that the burette reading may be interpreted directly as percentage of alkali, when a particular quantity of sample is employed.

8.3.1 *Reagents*

8.3.1.1 *Standard hydrochloric acid or standard sulphuric acid* — 2·5 N, accurately prepared.

8.3.1.2 *Phenolphthalein indicator solution* — prepare as given in **8.2.1.2**.

8.3.2 *Procedure* — Measure exactly 10 ml of the detergent solution to be tested in an Erlenmeyer flask, add 5 drops of phenolphthalein indicator solution. From a burette graduated in tenths, add the standard hydrochloric acid or sulphuric acid, drop by drop, until the solution becomes colourless. The number of millilitres used equals percentage of alkali, calculated as sodium hydroxide (NaOH).

8.4 Laboratory Method — The method can be used to estimate either caustic alkali due to NaOH and part of carbonate and phosphate, or total alkali due to NaOH, all the carbonate and two-thirds of the phosphate, by using two indicators.

8.4.1 *Reagents*

8.4.1.1 *Standard sulphuric acid* — 0·1 N.

8.4.1.2 *Phenolphthalein indicator solution* — prepare as given in **8.2.1.2**.

8.4.1.3 *Methyl orange indicator solution* — Dissolve 0·05 g methyl orange in 100 ml of water.

8.4.2 *Procedure* — Transfer 5 ml of the test solution to a 250-ml graduated flask, and dilute to the mark with distilled water. Mix well, and transfer 50 ml of the solution to an Erlenmeyer flask. Add a few drops of phenolphthalein indicator solution and titrate with the standard sulphuric acid to a colourless solution. Note the volume of acid used (A). To the same solution add now a few drops of methyl orange indicator solution and continue to titrate until a slight pink colour appears. Note the volume of acid used for this second titration (B) (as NaOH).

8.4.3 *Calculation*

Percent by weight of caustic alkali due to NaOH and part of carbonate and phosphate (expressed as NaOH) $= A \times 0{\cdot}4$

Percent by weight of total alkali due to NaOH, all the carbonate and two-thirds of the phosphate (expressed as NaOH) $= (A + B) \times 0{\cdot}4$

9. QUANTITATIVE ANALYSIS OF MIXTURES OF DETERGENTS

9.1 Sodium Hydroxide and Sodium Carbonate Solution — If the mixture contains only sodium hydroxide and sodium carbonate, the following procedure shall be followed.

9.1.1 *Reagents*

9.1.1.1 *Standard hydrochloric acid* — 0·1 N.

9.1.1.2 *Phenolphthalein indicator solution* — prepare as given under **8.2.1.2**.

9.1.1.3 *Methyl orange indicator solution* — prepare as given under **8.4.1.3**.

9.1.2 *Procedure* — Weigh accurately one gram of the detergent and make up with water in a graduated flask to 100 ml. Pipette 10 ml in an Erlenmeyer flask and titrate with the standard hydrochloric acid using 3 drops of phenolphthalein as indicator. Note the volume of hydrochloric acid used. Then add 3 drops of methyl orange indicator solution and titrate until a pink colour appears.

9.1.3 *Calculation*

Sodium hydroxide (NaOH), percent by weight $= 4 (2V_1 - V_2)$

Sodium carbonate (Na_2CO_3), percent by weight $= 10{\cdot}6 (V_2 - V_1)$

where

V_1 = volume in ml of the standard hydrochloric acid used up in the first titration, and

V_2 = total volume in ml of the standard hydrochloric acid used up in both the titrations.

NOTE — If phosphates and/or silicates are present, this simple titration method is not used. Titration to the phenolphthalein end-point will then give all the sodium hydroxide and metasilicate, one-half of the carbonate and one-third of the trisodium phosphate. If the silicate and phosphate are estimated separately (*see* **9.2** and *see* **9.3**), it is possible to correct the titration values but as the chances of error are increased, it is advisable also to determine the carbonate separately.

9.2 Silicates

9.2.1 *Reagents*

9.2.1.1 *Hydrochloric acid* — 1 : 1.

9.2.1.2 *Concentrated hydrochloric acid* — sp gr 1·16

9.2.1.3 *Dilute hydrochloric acid* — 1 : 20.

9.2.2 *Procedure* — Weigh accurately about 5 g of the detergent mixture and heat in a crucible for 15 minutes at about 600° ± 20°C to destroy organic matter. Transfer to a 10-cm porcelain dish, add 25 ml of hydrochloric acid (1 : 1) and stir to make homogeneous. Evaporate to dryness on a water-bath with constant stirring. Then heat at 110° to 120°C for one hour. Moisten the residue with 5 ml concentrated hydrochloric acid, add 75 ml water and heat on a water-bath for 20 minutes to dissolve any salts. Filter through a No. 41 filter paper or its equivalent and wash the precipitate with dilute hydrochloric acid. Then wash with distilled water until free from chlorides (no turbidity with silver nitrate). Return the filtrate to the original dish, evaporate, dry, acidify and filter as described above. Ignite both the papers at 1 000° ± 20°C in a weighed platinum crucible to constant weight. This gives crude silica which is sufficiently accurate for the purpose.

9.2.3 *Calculation*

$$\text{Sodium metasilicate, percent by weight} = 203 \frac{W_1}{W_2}$$

where

W_1 = weight in g of crude silica, and

W_2 = weight in g of the detergent taken for the test.

9.3 Phosphates

9.3.1 *Procedure* — Determine phosphorus volumetrically according to the method described under **19** of IS : 1479 (Part II)-1961 on the filtrate (*see* **9.2.2**) or an aliquot of it.

9.3.2 *Calculation*

a) Phosphorus pentoxide (P_2O_5), percent by weight $= 2.292\ W = W_1$

where

W = the percentage of phosphorus as calculated according to **19** of IS : 1479 (Part II)-1961.

b) Trisodium phosphate (Na_3PO_4), percent by weight $= 2.31\ W_1$

NOTE — This method gives the total phosphate. If polyphosphates are present, these shall be determined separately and the difference then gives the simple alkaline phosphates.

9.4 Sodium Hexametaphosphate and Other Polyphosphates — There are a number of polyphosphates possessing the power to sequester calcium salts and so soften water. The best known is called sodium hexametaphosphate and the formula is often given as $Na_6(PO_3)_6$ but the constitution is still a matter of controversy.

For the estimation of polyphosphates, as distinct from the simple meta, ortho and pyrophosphates, the following method may be used. Detergents, if at all they contain polyphosphates, usually contain about 5 percent concentration of the total solid detergent. A solution containing about 0·1 percent polyphosphate (*w/v*) is required, so usually a 2 percent solution of the detergent is convenient to use.

9.4.1 *Reagents*

9.4.1.1 *Methyl red indicator solution* — 0·5 g in 100 ml of alcohol, 95 percent (*v/v*).

9.4.1.2 *Standard hydrochloric acid* — 1 N.

9.4.1.3 *Barium chloride solution* — 10 percent (*w/v*) in water.

9.4.1.4 *Barium chloride* — one percent (*w/v*) in water.

9.4.1.5 *Dilute nitric acid* — 1 : 1 (*v/v*).

9.4.1.6 Other reagents given under **19** of IS : 1479 (Part II)-1961.

9.4.2 *Procedure* — Weigh 2 g of the detergent accurately and make up to 100 ml with water in a 100-ml graduated flask. Pipette 50 ml of the solution in a 250-ml beaker, add 1 ml of methyl red indicator solution and titrate with standard hydrochloric acid until the indicator changes to red, then add a further quantity of 0·5 ml of the standard hydrochloric acid. Add slowly, with constant stirring, 15 ml of 10 percent barium chloride solution and allow the precipitate to settle. When clear, add a few drops of barium chloride solution to ensure that the precipitation is complete, adding further amounts if necessary. Finally filter through a No. 3 Whatman filter paper or its equivalent and wash the precipitate with one percent barium chloride solution. Dissolve the precipitate in dilute nitric acid and wash through into a conical flask. Add a further 50 ml of dilute nitric acid and make up with water to 100 ml. Boil gently for 15 minutes, cool to 40° to 45°C and add 50 ml of molybdate solution at the same temperature. The phosphate which had previously been present as polyphosphate is now present as phosphomolybdate.

9.4.2.1 Place the solution in a shaking or stirring apparatus and shake or stir for 30 minutes at room temperature. Decant at once through a filter paper (Whatman No. 42 or its equivalent) and wash the precipitate twice by decantation with 25- to 30-ml portions of water containing 2 percent of sodium nitrate, agitating thoroughly, and allow to settle. Transfer the precipitate to the filter and wash with cold water containing 2 percent of sodium nitrate until the filtrate yields a pink colour upon the addition of phenolphthalein indicator and one drop of standard alkali. Transfer the precipitate and the filter paper to a beaker. Dissolve the precipitate in a small excess of the standard alkali, add a few drops of phenolphthalein indicator and titrate with the standard hydrochloric acid. One millilitre of the standard sodium or potassium hydroxide solution = 0·436 4 mg of phosphorus.

9.4.3 *Calculation*

$$\text{Phosphorus, percent by weight} = \frac{0{\cdot}436\,4\,(V_1 - V_2)}{2\,W_1}$$

where

V_1 = volume in ml of the standard alkali used to dissolve the precipitate,

V_2 = volume in ml of the standard hydrochloric acid solution required for neutralization of excess alkali, and

W_1 = weight in g of the detergent taken for the test.

9.5 Carbonates — These may be most conveniently determined by addition of excess acid and removing the carbon dioxide evolved, the difference in weights being equivalent to the carbon dioxide. Moisture shall be trapped by a calcium chloride tube. An alternative method is to absorb the carbon dioxide evolved in potassium hydroxide and soda-lime tubes.

9.5.1 *Apparatus* — Potash and soda-lime tubes attached to the side arm of a flask fitted with a tight rubber stopper through which passes a dropping funnel with a stopcock.

9.5.2 *Reagents* — Hydrochloric acid — 1 : 1.

9.5.3 *Procedure* — Sweep out the apparatus with a current of carbon-dioxide-free air and weigh the potassium hydroxide and soda-lime tubes. Introduce an accurately weighed quantity of the detergent (1 to 5 g) to the flask. Add an excess of hydrochloric acid, taking precautions to exclude carbon dioxide. Bring the solution to boil and keep gently boiling for 20 minutes, maintaining throughout a current of carbon-dioxide-free air. If sulphite is present, include also a tube of iodine solution before the first absorption tube. Note the increase in weight of the absorption tubes.

9.5.4 *Calculation*

a) Sodium carbonate, percent by weight $= \dfrac{240{\cdot}9\, w}{W}$

b) Sodium bicarbonate, percent by weight $= \dfrac{190{\cdot}9\, w}{W}$

where

w = weight in g of carbon dioxide absorbed in the absorption tubes, and

W = weight in g of detergent taken for the test.

9.6 Sulphites — Detergents for can washing and plant cleaning may contain sodium sulphite which protects tin from corrosion. It may be estimated in the following manner.

9.6.1 *Reagents*

9.6.1.1 *Standard iodine solution* — 0·1 N.

9.6.1.2 *Acetic acid* — glacial.

9.6.1.3 *Starch indicator solution* — Make a paste of 0·5 g of soluble starch in cold water, pour into 100 ml of boiling water, boil for 5 minutes, cool and bottle. The solution should be prepared afresh every two or three days.

9.6.1.4 *Standard sodium thiosulphate solution* — 0·1 N, standardized against potassium dichromate.

9.6.2 *Procedure* — Weigh accurately about 1 g of the detergent and add 100 ml of water to it. Add 10 ml of iodine solution slowly with stirring. Add 5 ml of glacial acetic acid. Titrate the excess iodine with the standard sodium thiosulphate solution, adding the starch indicator solution when the solution is pale yellow in colour. Carry out a blank test with 10 ml of distilled water.

9.6.3 *Calculation*

$$\text{Sodium sulphite, percent by weight} = \frac{12{\cdot}6\,(V_1 - V_2)\,N}{W}$$

where

V_1 = volume in ml of the standard sodium thiosulphate solution used for the blank,

V_2 = volume in ml of the standard sodium thiosulphate solution used for the test,

N = the normality of the standard sodium thiosulphate solution, and

W = the weight in g of the detergent taken for the test.

9.7 Sulphates — Sodium sulphate when present as a constituent of detergents, as it commonly is with certain types, may be estimated by the following simple method which is most suitable for the synthetic type of detergent.

9.7.1 *Reagents*

9.7.1.1 *Bromophenol blue indicator solution* — 0·1 percent (*w/v*) in rectified spirit.

9.7.1.2 *Sulphuric acid* — 0·5 N.

9.7.1.3 *Phenolphthalein indicator solution* — prepare as in **8.2.1.2**.

9.7.1.4 *Sodium hydroxide solution* — 0·5 N.

9.7.1.5 *Ethyl alcohol* — 95 percent (*v/v*) neutral.

9.7.2 *Procedure* — Weigh a quantity of the detergent sufficient to yield about 200 mg of sodium sulphate in a 400-ml beaker. Thoroughly mix with 5 ml of water. using a glass rod. Add one drop of bromophenol blue indicator solution and neutralize with sulphuric acid. Raise the solution rapidly to nearly boiling (less than one minute), add one drop of phenolphthalein solution and make the solution just alkaline with one or two drops of sodium hydroxide. Then add 200 ml neutral ethyl alcohol and boil the solution gently for 30 minutes, the beaker being covered by a clock-glass. After cooling to about 50°C, filter the mixture through a tared sintered-glass crucible and wash the precipitate with 100 ml of warm neutral ethyl alcohol. Dry at $100° \pm 1°C$ and cool the crucible in a desiccator and weigh. Dry for another 30 minutes, cool in a desiccator and weigh. Repeat the process until the difference between two successive weighings is less than 1 mg. Record the lowest weight.

9.7.3 *Calculation*

$$\text{Sodium sulphate, percent by weight} = 100 \frac{w}{W}$$

where

w = weight in g of the dried precipitate, and

W = weight in g of the material taken for the test.

9.8 Moisture — Practically all detergents contain moisture either as free water or as water of crystallization or both. Thus ordinary washing soda is $Na_2CO_3.10\,H_2O$ and in 100 kg of this there are 65 kg of water. In assessing the true worth of a detergent, allowance has to be made for the effloresces water content. Different methods of drying result in the removal of different amounts of moisture. Thus sodium carbonate gives up its water at room temperature but caustic soda is deliquescent and absorbs water. Moisture content, therefore, can only be expressed in terms of a standard method, for example, drying in vacuum at room temperature or at 70°C or heated at 100°C to constant weight. The absorption of atmospheric carbon dioxide by caustic soda is a further complication. The best method, therefore, is to determine the constituents separately, for it is these which control the properties and value of any detergent.

9.9 Interpretation of Results of Analysis — Although methods have been described above to give results in terms of sodium hydroxide (NaOH), sodium carbonate (Na_2CO_3), sodium metasilicate (Na_2SiO_3), trisodium phosphate (Na_3PO_4) and sodium hexametaphosphate [$Na_6(PO_3)_6$], it shall be remembered that once a mixture of these, or other similar salts, such as disodium phosphate (Na_2HPO_4), sodium orthosilicate (Na_4SiO_4), etc, is dissolved in water an equilibrium mixture is obtained, and it is impossible to say precisely what salts were originally present. Thus a mixture of disodium phosphate and sodium hydroxide would react as trisodium phosphate. However, this does not affect the issue as it is the final mixture which is the effective detergent and which is analysed.

10. CHLORINE IN STERILIZING SOLUTIONS

10.1 Sterilization of dairy equipment by hypochlorite or chloramine solutions is not recommended except in cases of difficulty with heat resisting organisms which normal heat sterilization will not destroy. In such cases, where facilities for heat sterilization do not exist or where the shape or size of the equipment make the use of steam or hot water impracticable, hypochlorite sterilization may be used. The solution should be at a strength not greater than 200 parts per million (ppm) or for maximum bactericidal effect not less than 100 ppm of available chlorine for a contact time of between 10 and 15 minutes.

10.1.1 During the washing of cans and bottles, the strength of the chlorine solution shall be checked in the beginning, the middle and at the end of the day's operation. Sterilizing solutions used for flushing or spraying contact surfaces of milk equipment shall be tested daily immediately before use and immediately after it has contacted the milk equipment surfaces. The chlorine in sterilizing solution may be estimated rapidly by colorimetric method (*see* **10.2**), or by titration (*see* **10.3**).

10.2 Colorimetric Method — The method depends on the depth of yellow colour produced when an orthotoluidine reagent is added to a relatively dilute solution of sodium hypochlorite. The colour produced is fairly proportional to the amount of chlorine present provided no other oxidizing substance is present.

10.2.1 Ready-made colour discs for different strengths of chlorine and indicator solutions are available for use with a comparator. Free chlorine in drinking water is also estimated by the same method.

10.3 Laboratory Method

10.3.1 *Reagents*

10.3.1.1 *Standard sodium thiosulphate solution* — 0·1 N, accurately prepared.

10.3.1.2 *Acetic acid* — glacial.

10.3.1.3 *Potassium iodide* — powdered.

10.3.1.4 *Starch indicator solution* — prepare as given in **9.6.1.3**.

10.3.2 *Procedure* — Place 50 ml or a suitable aliquot of solution to be tested in a 250-ml Erlenmeyer flask. Add 1 to 2 g of potassium iodide and allow to dissolve. Then add 10 ml of glacial acetic acid and mix. Titrate the solution with 0·1 N sodium thiosulphate until the yellow colour almost disappears. Add 2 ml of starch indicator solution which gives blue colour with free iodine. Continue the titration till the blue colour is discharged.

10.3.3 *Calculation* — When 50 ml of solution is used, the available chlorine in parts per million is obtained by multiplying the number of millilitres of 0·1 N sodium thiosulphate used by 71.

11. TESTING OF BRINE SOLUTIONS

11.1 Solutions of calcium or sodium chloride are utilized as refrigerating brines, and calcium chloride brine is by far the most widely used in the dairy industry. If not properly managed and controlled, brine solutions may become corrosive to pipes and other metal parts in refrigerating systems. The rate of corrosion depends on *p*H of brine which is influenced by such factors as impurities (acidity, ammonia), density, concentration of dissolved oxygen and temperature. Brines are alkaline when freshly made (7·5 to 8·5) but due to the gradual absorption of carbon dioxide upon exposure to air, these solutions tend to become acid in reaction. In general, the more concentrated the brine solution, the lower the oxygen solubility and less the corrosive effect. Corrosion of metal by refrigerating brines may be retarded by holding oxygen concentration to a minimum, by maintaining an alkaline reaction and by treatment with corrosion inhibitor (chromites, chromic acid, or chrom-gluconates). Brines should preferably be tested monthly for specific gravity (sp gr 1·200 to 1·250 at 15·5°C corresponding to 22 to 27 percent by weight of calcium chloride), chemical reaction (*p*H about 8·0), and percent of corrosion inhibitor (0·14 to 0·18 percent as sodium bichromate) and for possible presence of free ammonia. Filter the brine solution before testing.

11.1.1 *Alkalinity* — Take about 3 ml of the clear brine in a test-tube and add a few drops of phenolphthalein indicator solution (*see* **8.2.1.2**). Mix. If the resultant solution has a faint pink colour, then the reaction of the brine is correct, that is at a *p*H of about 8·4; if the colour is a deep red, the brine is too strongly alkaline; if the solution is colourless, the brine has an acid reaction.

11.1.2 *p*H — This is determined rapidly by using a universal indicator. To 10 ml of brine, add one drop of universal indicator. Mix and observe the colour. Assess the approximate *p*H as follows:

Reaction	*Colour*	*Approximate pH*
Acid	Orange-red	5·0
	Orange	5·5
	Orange-yellow	6·0
	Yellow	6·5
	Neutral greenish yellow	7·0 to 7·5
Alkaline	Green	8·0
	Bluish-green	8·5
	Greenish-blue	9·5
	Blue	9·5
	Violet	10·0

The correct colour is a bluish-green indicating *p*H of about 8·4.

11.1.3 *Determination of Alkalinity* — For adjusting the correct alkalinity (*p*H) of the brine, the reaction of brine shall be determined fairly accurately with the help of a *p*H meter.

11.1.3.1 *Apparatus*

a) *pH meter* — with glass and calomel electrodes.

b) *Stirrer* — preferably motor-driven.

c) *Burette* — 25 ml, graduated in 0·1 ml.

11.1.3.2 *Reagents*

a) *Buffer solution* — for standardizing the *p*H meter.

b) *Standard hydrochloric acid* — 0·1 N.

c) *Standard sodium hydroxide* — 0·1 N.

11.1.3.3 *Procedure* — Standardize the *p*H meter against buffer solutions following exactly the instructions given with the instrument.

Place 100 ml of filtered brine in a 150-ml beaker and determine the *p*H.

With the brine solution in constant agitation, adjust the *p*H to 8·0 by slowly adding from the burette a measured quantity of 0·1 N standard sodium hydroxide solution or 0·1 N standard hydrochloric acid, depending upon whether the initial *p*H was below or above 8·0.

Record the quantity required to adjust the *p*H.

11.1.3.4 *Adjustment of pH* — A *p*H value of approximately 8·0 may be regarded as satisfacotry. Values above or below 8·0 indicate a need for adjustment.

a) When the brine is acidic, sodium hydroxide must be added. The amount of sodium hydroxide to be added when 1 ml of 0·1 N sodium hydroxide is required to adjust the *p*H to 8·0 is 4·0 g for 100 litres of brine.

b) When the brine is alkaline, sodium dicromate ($Na_2Cr_2O_7.2H_2O$) is added to increase the *p*H and to inhibit corrosion. Hydrochloric acid shall not be used to reduce the *p*H. The amount of sodium dichromate to be added when 1 ml of 0·1 N standard hydrochloric acid is required to bring the *p*H to 8·0 is 12·3 g for 100 litres of brine.

11.1.4 *Determination of Corrosion Inhibitor* — The most commonly used inhibitor for calcium chloride is sodium dichromate. The following laboratory method is used to determine the percentage of inhibitor in terms of sodium dichromate.

11.1.4.1 *Reagents*

a) *Sodium thiosulphate* — 0·1 N.

b) *Starch indicator solution* — prepare as in **9.6.1.3**.

c) *Potassium iodide* — 15 percent (*w/v*) (prepare fresh).

d) *Concentrated hydrochloric acid* — sp gr 1·16.

11.1.4.2 *Procedure* — Weigh 50 g of the filtered brine into 250-ml Erlenmeyer flask. Add 5 ml of potassium iodide solution, and then 5 ml of concentrated hydrochloric acid. Mix and titrate against the standard sodium thiosulphate solution till brown colour nearly disappears. Add 1 ml of starch indicator solution, mix, and continue titration till the colour changes from dark blue to green.

11.1.4.3 *Calculation*

$$\text{Sodium dichromate } (Na_2Cr_2O_7.2H_2O), \text{ percent by weight} = \frac{4{\cdot}976\,5\, VN}{W}$$

where

V = the volume in ml of the standard sodium thiosulphate used for titration,

N = the normality of the standard sodium thiosulphate solution, and

W = the weight in g of filtered brine taken for the test.

11.1.4.4 *Interpretation* — The concentration of corrosion inhibitor (as sodium dichromate) should be maintained at 0·14 to 0·18 percent.

12. DETECTION OF AMMONIA IN BRINE

12.1 If ammonia gas is used as a refrigerant, it may leak into the brine.

12.2 If the brine smells of ammonia, or if the smell of ammonia is obtained when a few drops of sodium hydroxide solution are added to some brine taken in a test-tube, a large amount of ammonia is present in the brine.

12.3 To detect small quantities of ammonia, take some of the brine in a test-tube, add a few drops of sodium hydroxide solution and heat. Hold moistened red litmus paper in the vapour evolved. If the litmus paper turns blue, the presence of ammonia is indicated.

13. DETECTION OF LEAKS IN REFRIGERATION PIPES

13.1 Detection of Ammonia — Leaks in ammonia pipes shall be tested from time to time by the following tests:

a) Smear all pipe joints liberally with soap suds. Leaks will be indicated by the formation of bubbles.

b) Pass a piece of moistened red litmus paper around the joint. If the litmus paper turns blue, it indicates a leak.

c) Pass a lighted sulphur stick around the joints. If a dense cloudy vapour is formed, it indicates a leak.

NOTE — Sulphur sticks are made by dipping small pine or wood sticks in melted sulphur several times to build up a layer of sulphur on one end of the stick.

13.2 Detection of Freon 12, Methyl Chloride (Carrene 1) and Ethyl Chloride Leaks

13.2.1 Smear all pipe joints liberally with soap suds. Leaks will be indicated by the formation of bubbles.

13.2.2 To test for leak of freon, carrene or ethyl chloride, an alcohol-halide torch with a long air intake tube is used. With this alcohol-halide torch ignited, pass the intake tube near the different joints. If gas is leaking it will go through the intake tube to the flame and give forth a brilliant green hue, which is an indication of gas leak. The halide torch should be filled with alcohol and pumped in a room in which there is no possible chance of gaseous refrigerant being present; as otherwise, pumping up of refrigerant from the atmosphere into the torch would give a false indication of the presence of a leak. The alcohol used shall be exceptionally clean as the liquid passages are very small. To light one of these torches, the flame chamber must be preheated and the air intake tube opening stopped with one finger until the flame is burning well.

14. DETECTION OF LEAKS THROUGH PIN-HOLES IN PASTEURIZERS

14.1 General — Pin-holes in pasteurizer plates are likely to be developed due to corrosion of the stainless steel by brine or unsuitable cleaning agents. The plates should only be checked for pin-holes after a thorough check of the pasteurization plant record and control system. The commonest causes of a phosphatase positive test on pasteurized milk are underheating, not heating for the specified time and more direct methods of contamination of pasteurized milk with raw milk.

14.2 Reagents — Prepare a 2 percent paste of soluble starch by boiling with water for 30 minutes. To 100 ml of this solution add 1 ml of one percent phenolphthalein indicator solution neutralized with dilute (approximately 0·1 N) hydrochloric acid till colourless.

14.3 Procedure — Paint on the milk side of the pasteurizing plates a thin layer of the starch reagent and allow to dry, keeping the plates horizontal. Circulate through the water side a weak solution of sodium hydroxide (0·1 N).

14.4 Interpretation — Plates will show a pinkish area on the layer on starched plates in case pin-holes are present.

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